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Introduction

Humor generation is one of the clearest examples of a language task that matters precisely because it is hard to verify automatically. A joke may be grammatically correct, semantically coherent, and even topically relevant while still failing at the central target: making a reader laugh. This makes humor generation a useful stress test for modern language-model post-training. In domains such as mathematics, coding, or formal logic, reinforcement learning can rely on explicit verification signals. In humor, by contrast, the output space is open-ended, the number of acceptable answers is large, and human judgments are noisy. The same prompt can admit many good jokes, while a dataset usually records only one or a few. These properties make humor an informative case for studying reasoning in non-verifiable domains.

The relevance of the topic is determined by three converging trends. First, recent surveys show that computational humor is moving from isolated rule-based or template-based systems toward large language model pipelines, yet the gap between fluent text and genuinely funny output remains substantial [Loakman et al., 2025a]. Second, post-training research increasingly suggests that supervised fine-tuning (SFT) and reinforcement learning optimize different capabilities: SFT is effective at imitation, whereas reinforcement learning is more promising when generalization beyond the training surface is required [Chu et al., 2025]. Third, reasoning-focused reinforcement learning has recently produced strong results in domains with explicit verifiers, motivating the question of whether similar methods can be adapted to creative tasks where direct verification is impossible [DeepSeek-AI, 2025, Zhou et al., 2025, Yu et al., 2025].

This thesis studies humor generation as a non-verifiable reasoning problem. The central idea is that humor does not only require a good final answer; it often benefits from exploratory reasoning over premises, associations, contrasts, and punchline formats. Multi-step humor systems already exploit this fact at inference time by explicitly decomposing the creative process into intermediate

stages [Tikhonov and Shtykovskiy, 2024, Zhang et al., 2025, Kim and Chilton, 2025]. However, such systems are architecturally restrictive: the number of stages, the reasoning style, and the intermediate control signals are designed manually. The alternative investigated in this thesis is to construct a training setting in which a model can learn how to reason for humor generation without requiring a symbolic verifier.

The object of the study is the post-training of large language models for creative text generation. The subject of the study is the design of a multi-reference verifier-free reinforcement learning objective for humor generation together with the supporting data pipeline needed to make such an objective operational.

The goal of the thesis is to develop a preliminary methodological and software foundation for training humor-generation models with verifier-free reinforcement learning, while remaining faithful to the actual implementation status of the project. In practical terms, the work aims to transform a large joke corpus into prompt-reference sets suitable for multi-reference training and evaluation, formulate the corresponding reinforcement learning objective, and analyze whether the constructed data artifacts are adequate for a later MRVF training stage.

To achieve this goal, the following tasks are addressed:

1. analyze recent literature on computational humor, post-training, and verifier-free reasoning methods;

2. define the research gap between current humor pipelines and reinforcement learning for non-verifiable domains;

3. build a corpus-processing pipeline that converts raw jokes into weak prompts with multiple retrieved references;

4. implement candidate-generation and evaluation components for later benchmark comparisons;

5. formulate a multi-reference extension of verifier-free reinforcement learning and discuss its statistical consequences under incomplete references;

6. evaluate the current pilot data pipeline using descriptive statistics, qualitative examples, and implementation analysis;

7. outline the next experimental stage needed to turn the pilot into a full thesis-level training study.

The working hypothesis of the thesis is that humor generation can be made more amenable to reasoning-oriented post-training if the learning problem is reframed from "predict the single gold joke" to "assign probability mass to a set of acceptable jokes conditioned on a weak prompt." Under this hypothesis, the main obstacle is not only reward design, but also dataset structure: a useful training signal requires prompts that admit multiple valid references rather than isolated one-to-one joke mappings.

The scientific novelty of the work is threefold. First, it adapts the verifier-free reinforcement learning perspective of Zhou et al. [2025] and Yu et al. [2025] to a creative task where multiple valid outputs are expected by definition. Second, it makes explicit the difference between the original conceptual report and the current implemented system: rather than a full generative prompt inversion module, the present codebase realizes a retrieval-based approximation built on keywords, embedding search, and reference-set construction. Third, it provides an empirical pilot analysis of this approximation on real data, which is necessary before a more expensive reinforcement-learning stage can be justified.

The practical value of the work lies in the implemented software pipeline. The repository currently contains modular Python components for corpus collection, embedding generation, keyword extraction, reference retrieval, candidate generation, and pairwise evaluation. These components produce reusable parquet datasets and isolate stages that can later be reused for baseline generation, preference comparison, and reinforcement-learning experiments.

The empirical basis of the work consists of the combined joke corpus built from the Short Jokes and rJokes collections. In the current local state of the repository, the complete raw corpus contains 664,048 jokes, while the downstream pilot pipeline processes a smaller subset for prompt construction and reference analysis. This distinction is important: the thesis is preliminary not only in conceptual scope, but also in the current scale of the implemented experiments.

Methodologically, the thesis combines literature review, software engineering, corpus construction, embedding-based retrieval, mathematical derivation of an objective function, and descriptive analysis of intermediate artifacts. This combination reflects the actual status of the research problem: before large-scale

reinforcement learning can be defended, the data representation, prompt formulation, and reference structure must first be stabilized.

The structure of the thesis follows the standard university requirements. Chapter 1 reviews the literature on humor generation, evaluation, and reasoning-oriented post-training. Chapter 2 presents the methodology, including the implemented pipeline and the formal MRVF objective. Chapter 3 reports the current pilot results, discusses strengths and limitations of the constructed dataset, and explains what remains to be implemented for the final thesis. The work concludes with a summary of findings, followed by the list of references and appendices.

Chapter 1

Literature Review

1.1 Computational humor as a difficult NLP problem

Humor has long been treated as a hard case for natural language processing because successful jokes depend on more than grammatical well-formedness. They also depend on incongruity, shared knowledge, timing, and audience expectations. Early reviews already described humor as one of the more complex targets for artificial intelligence, noting that implemented systems were largely confined to narrow punning mechanisms and that broader humor processing would require substantial world knowledge and reasoning [Ritchie, 2001]. Two later surveys reach a similar conclusion from different angles. Amin and Burghardt [2020] show that computational humor generation has historically traded breadth for control: template- and rule-based systems could operationalize identifiable humorous mechanisms, but only within highly restricted domains. More recently, Loakman et al. [2025a] and Lemmens and De Marez [2026] argue that the large-language-model era has broadened coverage while leaving the central problem unresolved, since fluent text is not equivalent to humor that readers actually find funny.

A further difficulty is theoretical pluralism. Humor theory offers several recurring explanatory families—most notably incongruity-based accounts, but also superiority and relief traditions—yet no single framework exhausts the phenomenon. This matters computationally because different joke types rely on different mechanisms. Kao et al. [2016], for instance, formalize incongruity in puns as ambiguity plus distinctiveness and show that these variables predict human funniness judgments. Their work is important not only because it operationalizes a classical humor theory, but also because it illustrates the limits of narrow

formalization: a model that explains homophonic puns does not thereby explain narrative jokes, topical jokes, or socially situated humor. Recent theory-driven work in humor detection starts from the same premise, namely that no single comprehensive humor theory is sufficient and that computational models typically operationalize only fragments of humor [De Marez et al., 2024]. For the present thesis, the consequence is methodological: humor generation should be treated as a multi-mechanism creative task rather than as a single scalar attribute.

1.2 From narrow generators to LLM-era humor generation

Before the rise of large language models, humor generation systems were usually narrow by design. As summarized by Amin and Burghardt [2020], earlier work often relied on hand-written templates, lexical resources, phonetic transformations, or tightly constrained generation procedures. These systems had an important advantage: they encoded at least one explicit humorous mechanism. Their disadvantage was low adaptability. They produced humor only under carefully chosen linguistic conditions and therefore did not scale well beyond punning, stock joke forms, or limited domains.

Large language models invert that trade-off. They can produce open-ended humorous text over a far broader range of prompts, but fluency does not guarantee robust humor competence. This is visible in recent large-scale evaluations. Zhang et al. [2024] introduce a multimodal cartoon-captioning benchmark built from more than 250 million human ratings over more than 2.2 million captions, and show that current fine-tuning methods remain limited on creative tasks; even strong systems such as GPT-4 and Claude remain below top human contestants. Humor-specific understanding studies point in the same direction. Xu et al. [2024] show that large language models exhibit what they call "lazy pun generation," which is one reason why stronger evaluation metrics are needed for pun understanding. Cocchieri et al. [2025] construct a contamination-resistant humor benchmark and find that most language models still underperform humans on pun comprehension, resolution, and generation. Going beyond puns, Loakman et al. [2025b] show that current models, including reasoning models, do not reliably explain all joke types, which highlights how much of the existing literature

remains focused on relatively simple joke forms.

Taken together, these studies suggest that current models are better at reproducing the local surface form of familiar jokes than at generalizing across humor mechanisms. This distinction is important for the present thesis. A system can be good at producing joke-like text without being good at humor in the stronger sense required by open-ended generation.

1.3 Humor evaluation is multidimensional and judge-dependent

Evaluation is a central difficulty in computational humor. Humor is subjective, but the problem is not only subjectivity in the colloquial sense. What counts as a good joke depends on audience, context, delivery, and the dimension being evaluated. Even dataset papers emphasize that humor varies across topic, time, and recipient, making it difficult to identify a single target variable [Weller and Seppi, 2020]. For this reason, recent humor work increasingly avoids treating funniness as a single scalar judgment.

The clearest example is Sakabe et al. [2025], who evaluate Oogiri responses on six dimensions: novelty, clarity, relevance, intelligence, empathy, and overall funniness. Their results show that current language models generate responses between low- and mid-tier human performance, but they notably lag in empathy; the authors further argue that this deficit helps explain why model-based evaluation does not reproduce human humor assessment well. The significance of this result is twofold. First, it shows that humor quality is not reducible to "is this funny?" Second, it shows that human and model judgments may privilege different qualities.

A complementary perspective comes from Winters and Van der Stockt [2025], who compare human comedians and GPT-4 in a live improvisational setting rather than in an offline text-only benchmark. Their study is valuable because it isolates a stronger comparison point: both humans and the model must respond rapidly to the same audience suggestions. Human jokes are still preferred slightly more often, but the margin is not overwhelming, and the design highlights how much delivery and situational context affect humor evaluation. This is a useful corrective to benchmark-only thinking. A text that performs well in static judging

is not necessarily the text that works best in real comedic interaction.

These findings also clarify the role of LLM-as-a-judge methods. Structured evaluation frameworks such as Hashemi et al. [2024] are useful because they replace one-dimensional scoring with explicit rubrics and calibration. However, that paper also shows that raw LLM rubric judgments do not automatically align well with human ratings. For humor generation, this implies that LLM judges are best treated as measurement instruments with a limited domain of validity, not as transparent replacements for human preference.

1.4 Reasoning and multi-stage generation for humor

One of the most important shifts in recent humor-generation research is the move from one-shot continuation to explicit reasoning or staged creative search. At the prompting level, Chen et al. [2023] model humor generation through step-by-step comedy-writing instructions for GPT-3. Zhong et al. [2024] go further by arguing that standard chain-of-thought prompting is poorly matched to creative tasks, and propose a Leap-of-Thought paradigm for Oogiri-style humor generation that emphasizes associative jumps rather than strictly sequential decomposition.

More explicit generation systems strengthen this line of work. Tikhonov and Shtykovskiy [2024] reconstruct one-liner joke writing as a multistep process and report improvements over zero-shot GPT-4 and other baselines in human evaluation. Kim and Chilton [2025] frame humor generation as the combination of cognitive, social, and creative skills; in their meme-captioning study, skill-augmented systems substantially outperform basic large-language-model baselines and approach top-rated human captions for the target audience. In multimodal settings, Zhang et al. [2025] propose a theory-guided multi-stage reasoning framework intended to improve both interpretability and humor quality.

These papers are important because they collectively reject the assumption that humor is best produced by a single left-to-right continuation. Instead, they model humor generation as a search process over premises, associations, reframings, and audience-sensitive choices. At the same time, they share a limitation that is directly relevant to the present thesis: the structure of the reasoning process is hand-specified. The number of stages, the type of intermediate representations,

and the order of operations are fixed by the researcher. Such systems are therefore strong demonstrations of the usefulness of reasoning, but weaker evidence that a model can learn for itself when and how to reason for humor.

1.5 Why standard post-training is a poor fit for humor

The most straightforward route to improving humor generation is supervised fine-tuning on large joke corpora. In practice, however, that objective is a poor match for the structure of the task. Joke corpora contain many repeated setups, familiar wordplay patterns, and conventional punchline forms. As a result, supervised fine-tuning can improve stylistic plausibility without necessarily improving creative generalization. Large humor resources such as rJokes are valuable for scale and topical variety, but they were not constructed as multi-reference generation datasets in which many distinct outputs can simultaneously be correct [Weller and Seppi, 2020].

Recent literature strengthens this concern from two directions. First, humor-specific work shows that even very large preference datasets do not make creative alignment easy. Zhang et al. [2024] explicitly highlight the limitations of RLHF- and DPO-style fine-tuning on creative captioning. Second, post-training work outside humor makes the memorization-generalization distinction sharper. Chu et al. [2025] show that reinforcement learning generalizes more effectively than supervised fine-tuning to unseen textual and visual variants, whereas supervised fine-tuning tends to memorize training patterns and struggle out of distribution. This contrast is especially consequential in humor generation, where a weak prompt can have many good continuations and where memorizing frequent joke shells is not the same as producing an apt and surprising punchline.

Preference optimization has its own difficulties. RLHF assumes that a reward model can learn a stable proxy for the target behavior, but humor judgments are noisy and heterogeneous. Gao et al. [2023] show more generally that aggressively optimizing an imperfect reward model can harm true performance. Humor-specific evaluation suggests that the risk is amplified in creative domains: Sakabe et al. [2025] show systematic divergence between human and model humor judgments, while Winters and Van der Stockt [2025] show that humor quality de-

pends on aspects of performance that are hard to capture in offline textual reward signals. For these reasons, neither supervised fine-tuning nor standard preference optimization cleanly solves the central problem.

1.6 Reasoning-oriented post-training and verifier-free reinforcement learning

The recent success of reasoning-oriented post-training has come mainly from domains with objective verification. Wei et al. [2022] show that intermediate reasoning can substantially change model behavior on tasks that reward multi-step inference. Lightman et al. [2023] extend this logic from prompting to supervision, showing that process supervision outperforms outcome-only supervision on the MATH benchmark. DeepSeek-AI [2025] then demonstrate that reinforcement learning can elicit long-form reasoning without relying on human-authored chains of thought, but the clearest gains are reported in verifiable domains such as mathematics, coding, and related STEM tasks.

This creates the central transfer problem for humor. Humor may also benefit from extended reasoning, yet there is usually no symbolic verifier that can declare a joke correct. Recent verifier-free approaches attempt to close this gap by replacing the explicit verifier with intrinsic probabilities assigned to reference answers. Zhou et al. [2025] propose VeriFree and show that, under a single-reference assumption, maximizing the conditional probability of the reference answer given the prompt and reasoning trace is scalar-equivalent to maximizing an exact-match verifier objective. Yu et al. [2025] develop a related approach and emphasize the need for variance control when token probabilities are used as rewards in general domains.

For humor, however, the single-reference assumption is not a minor technical detail. It is the wrong data model. Prompts such as "write a joke about atheism" or "write a pun using prophet" admit many plausible responses, and any dataset will observe only a small and biased subset of them. In other words, humor is both multi-reference and only partially observed. This means that verifier-free reasoning methods are conceptually relevant to humor generation, but not directly applicable without reformulating the reward around sets of references rather than single targets.

1.7 Research gap

The literature reviewed above leaves a precise gap. Humor-generation studies increasingly show that staged reasoning, skill decomposition, or theory-guided generation can improve outputs, but these systems mostly rely on hand-designed inference pipelines [Chen et al., 2023, Zhong et al., 2024, Tikhonov and Shtykovskiy, 2024, Kim and Chilton, 2025, Zhang et al., 2025]. Work on reasoning-oriented reinforcement learning shows that useful reasoning behaviors can emerge from optimization rather than from fully scripted chains, but the strongest existing formulations assume either objective verifiability or a single reference answer [Lightman et al., 2023, DeepSeek-AI, 2025, Zhou et al., 2025, Yu et al., 2025]. Meanwhile, humor-evaluation research shows that funniness is multidimensional, context-sensitive, and only imperfectly captured by model judges [Zhang et al., 2024, Sakabe et al., 2025, Winters and Van der Stockt, 2025, Hashemi et al., 2024].

The present thesis is positioned at the intersection of these findings. It takes seriously the claim that humor benefits from reasoning, but rejects the need to hard-code a single creative script. It also takes seriously the verifier-free reinforcement-learning literature, but argues that creative tasks require a multi-reference formulation because multiple outputs can be simultaneously good and the observed reference set is incomplete. The resulting research question is therefore narrower and more concrete than "can AI be funny?" The question is whether verifier-free reasoning objectives can be extended to partially observed multi-reference humor generation in a way that encourages creative search without collapsing onto memorized joke templates.

Chapter 2

Methodology

2.1 Research design and current implementation status

The methodology of this thesis is intentionally layered. The final target of the overall project is a trained humor-generation model optimized with a multi-reference verifier-free reinforcement learning objective. However, the current repository does not yet contain that final training stage. The file `src/pipelines/training.py` is still empty, while the surrounding infrastructure for data construction and evaluation is implemented. For that reason, the methodology chapter is divided into two interconnected parts:

1. the *implemented pilot pipeline*, which is already present in code and can be analyzed empirically;

2. the *formal MRVF training objective*, which is specified in mathematical form as the next research stage.

This distinction is essential for academic honesty. The thesis should not describe unimplemented reinforcement-learning experiments as completed work. Instead, it should show that the implemented pipeline produces the intermediate artifacts required for those experiments and that the theoretical objective is aligned with the structure of those artifacts.

At a high level, the implemented system proceeds through the following stages:

1. collect and normalize a large joke corpus;

2. generate text embeddings for jokes;

3. extract salient keywords from each joke using embedding-guided selection;
4. form weak prompts from one-keyword and two-keyword groups;
5. retrieve multiple candidate references for each prompt using a FAISS index over joke embeddings;
6. split the resulting prompt-reference dataset into train, validation, and test partitions;
7. generate model outputs for a given set of prompts;
8. evaluate those outputs via pairwise comparison and Bradley–Terry style aggregation.

This pipeline already embodies the operational core of the thesis: it transforms isolated jokes into a many-to-many training and evaluation structure. The current pilot therefore serves as a feasibility study for the later reinforcement-learning stage.

2.2 Source corpora and preprocessing

The empirical base of the project is a merged corpus built from two publicly available joke collections:

- Short Jokes, downloaded from the `short-jokes-dataset` repository;
- rJokes, downloaded from the `rJokesData` repository.

The `JokesPipeline` module downloads the raw files, converts them to parquet, preserves source metadata, and assigns global integer identifiers. In the current local repository state, the merged corpus contains 664,048 jokes in total. Table 2.1 summarizes the exact counts extracted from the local data artifacts.

The two sources are complementary. Short Jokes contains many compact one-liners, while rJokes includes a broader range of narrative formats and internet-style setups. Their combination is useful because the later reference-retrieval stage benefits from topical and stylistic redundancy: repeated motifs increase the chance that several different jokes can answer the same weak prompt.

Table 2.1: Source corpora in the local repository state

Source	Number of jokes	Share of total
Short Jokes	231,657	34.9%
rJokes	432,391	65.1%
Total	664,048	100.0%

The preprocessing stage is intentionally conservative. The implementation preserves the original text, source name, source filename, and source-local identifier. No aggressive normalization is performed beyond stripping empty rows, because excessive normalization would risk removing stylistic cues that are important for humor.

2.3 Embedding generation

The first implemented transformation after corpus assembly is dense embedding generation. The `EmbeddingsPipeline` uses the model `qwen/qwen3-embedding-8b` with 1,024-dimensional output vectors. Embeddings are generated asynchronously in batches and stored in parquet shards. This stage serves two roles:

- it supports later keyword selection by measuring similarity between a joke embedding and candidate keyword embeddings;
- it provides the vector index used for reference retrieval.

The current local state contains 10,005 embedded jokes. This number is much smaller than the full raw corpus size, which indicates that the pilot has so far been run on a limited subset. This is not a defect; it is a reminder that the current thesis is preliminary and reports a pilot implementation rather than a final large-scale experiment.

2.4 Keyword extraction as a prompt approximation

The original research note proposed an inversion procedure in which a language model would rewrite each joke into one or several plausible prompts. The current implementation takes a different, more engineering-oriented route. Instead of using a generative inversion model, it approximates weak prompts through salient keywords extracted from each joke.

This approximation is implemented in `KeywordsPipeline`. The procedure is as follows:

1. tokenize each joke into unigram candidates with a `CountVectorizer`;
2. remove stop words and retain at most 64 candidate tokens per joke;
3. embed each candidate token using the same embedding family used for joke embeddings;
4. compute cosine similarity between the joke embedding and each candidate embedding;
5. select the top 3 candidates using maximal marginal relevance (MMR) with diversity coefficient 0.7.

The logic behind this design is straightforward. A purely frequency-based keyword extractor would prefer common tokens that are not semantically central to the joke. A purely similarity-based selector would risk choosing near-duplicates. The MMR step balances relevance and diversity, so that the final keyword set is short but not redundant.

Formally, for a joke j with embedding $e(j)$ and a set of candidate keyword embeddings $\{e(k_1), \ldots, e(k_m)\}$, the relevance score is the cosine similarity

$$s_{\mathrm{rel}}(k_i, j) = \frac{e(k_i) \cdot e(j)}{\|e(k_i)\| \|e(j)\|}.$$

MMR then selects the next keyword by maximizing a trade-off between relevance and novelty relative to previously selected keywords.

The practical output of this stage is a short list of tokens or short phrases, usually of length three per joke in the current pilot. Across the 1,005 jokes that have already passed through this stage, the average number of retained keywords is 2.99, which means that nearly every joke yields a full three-keyword representation.

This stage deserves a careful interpretation. It does *not* recover the full latent intent of the joke. Instead, it provides a weak semantic handle that can later be expanded into reference sets. For a preliminary thesis, this compromise is acceptable because the immediate goal is not to solve prompt inversion perfectly, but to build a tractable multi-reference training substrate.

2.5 Prompt construction and reference retrieval

2.5.1 Weak prompts

Once keywords are extracted, the `ReferencesPipeline` creates one-keyword and two-keyword groups for each joke. If a joke produces the keyword set $K(x) = \{k_1, k_2, k_3\}$, the implementation generates all combinations whose size lies between the configured minimum and maximum:

$$\mathcal{G}(K) = \{\{k_1\}, \{k_2\}, \{k_3\}, \{k_1, k_2\}, \{k_1, k_3\}, \{k_2, k_3\}\}.$$

Each keyword group is rendered into a natural-language prompt using the template

```
Write a joke using the following keyword(s): {keywords}
```

This design is simple, but it creates a useful hierarchy of prompt specificity. One-keyword prompts are broad and ambiguous. Two-keyword prompts are narrower and better suited to retrieving semantically compatible references. The pilot intentionally excludes three-keyword prompts from retrieval because the resulting prompts would often be too specific and therefore less likely to admit multiple distinct answers.

2.5.2 Reference retrieval

The retrieval index is built over the embedded joke collection using FAISS with an IVF Flat inner-product index. The main configuration values used in the current pilot are:

- embedding model: `qwen/qwen3-embedding-8b`;
- dimensionality: 1,024;
- number of lists (`nlist`): 4,096;
- probe count (`nprobe`): 32;
- top-k references retained after masking: 5;
- minimum similarity threshold: 0.5;
- oversampling during search: 10 additional candidates.

For a prompt x derived from a keyword group, the system embeds the textual prompt, searches the FAISS index, filters low-similarity items, and keeps up to five nearest jokes as references. If the retrieved references are denoted $Y^*(x) = \{y_1, \ldots, y_n\}$, then in the pilot implementation n is bounded above by 5 and may be smaller or even zero after filtering.

The resulting dataset can be written as

$$\mathcal{D}_{\text{ref}} = \{(x_i, Y^*(x_i))\}_{i=1}^{N},$$

where x_i is a weak keyword-based prompt and $Y^*(x_i)$ is the set of retrieved joke references associated with it.

2.5.3 Deduplication and data splitting

An important post-processing step deduplicates rows by keyword group. This matters because different source jokes can generate the same weak prompt. Without deduplication, the final dataset would overcount prompt forms that happen to arise frequently in the seed subset.

After deduplication, the dataset is split into training, validation, and test sets with fractions 0.8, 0.1, and 0.1 respectively (implemented through successive train-test splits with a fixed random seed). The exact statistics of this pilot split are discussed in Chapter 3.

2.6 Candidate generation and evaluation infrastructure

Although the main training stage is not yet implemented, the repository already contains the components needed to benchmark future models once they exist.

2.6.1 Candidate generation

The `CandidatesPipeline` takes prompt rows from the reference dataset and asks a generation model to produce jokes. In the current configuration, the default generator is `gpt-4.1-mini` with temperature 1.0 and a cap of 128 completion tokens. The prompt is the same reference prompt used during retrieval:

```
Write a joke using the following keyword(s): {keywords}
```

This uniform prompting strategy is useful because it isolates the effect of the model rather than the effect of extensive prompt engineering. At the same time, it means that future results will need to be interpreted as a property of the model-and-prompt pair, not of the model alone.

2.6.2 Pairwise evaluation

The `EvaluationPipeline` compares model outputs pairwise for the same prompt. Each pair is judged by a language-model evaluator, again configured as `gpt-4.1-mini` in the current local setup. The system prompt asks the judge to choose which joke is funnier for a general audience and explicitly forbids ties. After collecting pairwise winners, the pipeline aggregates the results into a leaderboard using a Bradley–Terry style score estimation procedure.

This evaluation design matches the literature better than absolute rating scales. Humor is often easier to compare than to score in isolation. Pairwise

comparison reduces some scale-calibration problems and can later be combined with human annotation if the project moves beyond LLM judging.

The crucial limitation is that this evaluation infrastructure is implemented, but no committed evaluation artifacts are currently present in the repository. Therefore, the thesis can describe the methodology and planned use of the evaluator, but it cannot honestly claim completed benchmark results from this module yet.

2.7 Formalizing multi-reference verifier-free reinforcement learning

2.7.1 Notation

The formal part of the thesis follows the notation commonly used in reasoning-RL research. Let

- x denote the prompt;
- z denote a reasoning trace produced by the policy;
- y denote the final answer produced by the policy;
- y^* denote a single reference answer;
- $Y^*(x)$ denote a set of acceptable reference answers for prompt x.

The policy is factorized as

$$\pi_\theta(z, y \mid x) = \pi_\theta(z \mid x)\pi_\theta(y \mid x, z).$$

This factorization is conceptually useful even if the underlying language model generates a single text stream. It separates the reasoning trace from the final answer and clarifies which terms are being optimized by the reward.

2.7.2 RL with verifiable rewards

If a symbolic verifier exists and there is exactly one correct answer y^*, then the reward can be written as

$$R(y, y^*) = \mathbb{I}\{y = y^*\}.$$

The expected objective is

$$J_{\mathrm{RLVR}}(\theta \mid x, y^*) = \mathbb{E}_{z\sim\pi_\theta(z|x)}\mathbb{E}_{y\sim\pi_\theta(y|x,z)}[R(y, y^*)].$$

Using the score-function identity,

$$\nabla_\theta \mathbb{E}_{u\sim p_\theta(u)}[f(u)] = \mathbb{E}_{u\sim p_\theta(u)}\big[f(u)\nabla_\theta \log p_\theta(u)\big],$$

the policy gradient becomes

$$\nabla_\theta J_{\mathrm{RLVR}} = \mathbb{E}_{z,y}\Big[R(y, y^*)\big(\nabla_\theta \log \pi_\theta(z \mid x) + \nabla_\theta \log \pi_\theta(y \mid x, z)\big)\Big].$$

This expression is standard in reasoning-RL settings and is the point of departure for verifier-free reformulation.

2.7.3 VeriFree

Zhou et al. [2025] observe that under the single-reference assumption,

$$\mathbb{E}_{y\sim\pi_\theta(\cdot|x,z)}[\mathbb{I}\{y = y^*\}] = \pi_\theta(y^* \mid x, z).$$

Therefore, the scalar RLVR objective can be rewritten as

$$J_{\mathrm{VF}}(\theta \mid x, y^*) = \mathbb{E}_{z\sim\pi_\theta(z|x)}[\pi_\theta(y^* \mid x, z)].$$

The practical attraction of this objective is that it avoids explicit sampling of the final answer for reward computation. Instead, the model's conditional probability of the reference answer plays the role of the reward.

Differentiating this objective yields

$$\nabla_\theta J_{\mathrm{VF}} = \mathbb{E}_{z\sim\pi_\theta(z|x)}\Big[r_\theta(x,z,y^*)\big(\nabla_\theta \log \pi_\theta(z \mid x) + \nabla_\theta \log \pi_\theta(y^* \mid x,z)\big)\Big],$$

where

$$r_\theta(x,z,y^*) = \pi_\theta(y^* \mid x,z).$$

2.7.4 Why single-reference assumptions fail for humor

The single-reference assumption is inappropriate for creative tasks in general and for humor in particular. A prompt such as "write a joke about atheism" or "write a joke using the keyword ears" has many acceptable answers. If only one observed answer is rewarded, then the training signal treats all alternative good jokes as negatives. This is not a small technical inconvenience; it reverses the intended semantics of the task.

Let $Y^*(x) \subset \mathcal{Y}$ denote the set of known good references for prompt x. Then the natural verified reward becomes

$$R(y, Y^*) = \mathbb{I}\{y \in Y^*\}.$$

The corresponding multi-reference objective is

$$J_{\mathrm{MR}}(\theta \mid x, Y^*) = \mathbb{E}_z \mathbb{E}_{y\sim\pi_\theta(\cdot|x,z)}[\mathbb{I}\{y \in Y^*\}].$$

Since

$$\mathbb{E}_y[\mathbb{I}\{y \in Y^*\}] = \sum_{y\in Y^*} \pi_\theta(y \mid x,z),$$

the verifier-free reward can be generalized to

$$r_\theta(x,z,Y^*) := \sum_{y\in Y^*} \pi_\theta(y \mid x,z),$$

and the multi-reference verifier-free objective becomes

$$J_{\mathrm{MRVF}}(\theta \mid x, Y^*) = \mathbb{E}_{z\sim\pi_\theta(z|x)}[r_\theta(x,z,Y^*)].$$

This objective is the conceptual core of the thesis. It preserves the verifier-

free logic of Zhou et al. [2025] but replaces the single completion event with a union of acceptable completion events.

2.7.5 Incomplete reference sets

In humor generation, even the multi-reference set $Y^*(x)$ is still incomplete. The observed references for a prompt are only a subset of the jokes that could legitimately answer it. The repository's reference-construction pipeline makes this visible in practice: some prompts retrieve five strong references, some retrieve only one, and some retrieve none at all.

Suppose a subset $Y \subset Y^*$ is sampled uniformly without replacement and used as the observed reference set. Then the approximate reward is

$$r_\theta(x, z, Y) = \sum_{y \in Y} \pi_\theta(y \mid x, z).$$

If each member of Y^* has inclusion probability ρ, then

$$\mathbb{E}_Y[r_\theta(x, z, Y)] = \rho \sum_{y \in Y^*} \pi_\theta(y \mid x, z) = \rho\, r_\theta(x, z, Y^*).$$

Therefore the estimate is unbiased up to a constant scaling factor. This scaling is not fatal, especially if group-wise normalization is already used during optimization. The deeper problem is different: increasing mass on observed references also decreases mass on everything outside the observed set, which includes not only bad completions but also unseen good ones.

This false-negative risk motivates regularization. One natural choice is a Kullback–Leibler penalty relative to a reference policy π_{ref}:

$$\max_\theta \; \mathbb{E}_{x,z}[r_\theta(x, z, Y^*(x))] - \beta \mathbb{E}_x\big[\mathrm{KL}(\pi_\theta(\cdot \mid x) \| \pi_{\text{ref}}(\cdot \mid x))\big].$$

The KL term constrains catastrophic drift and makes it less likely that optimization will overfit the observed reference subset.

2.7.6 Training procedure as the next implementation stage

The intended training loop is not yet implemented in code, but it can be formulated clearly. For each prompt x, sample a group of reasoning traces

$$z_1, \ldots, z_G \sim \pi_\theta(z \mid x).$$

For each trace z_i, compute the reward

$$\hat{r}_i = r_\theta(x, z_i, Y^*(x)).$$

Then define a leave-one-out baseline

$$A_i = \hat{r}_i - \frac{1}{G-1} \sum_{j \neq i} \hat{r}_j,$$

which is unbiased for the score-function term involving z_i. The reasoning-gradient contribution becomes

$$g_z = \frac{1}{G} \sum_{i=1}^{G} A_i \nabla_\theta \log \pi_\theta(z_i \mid x),$$

while the reference-probability term is

$$g_r = \frac{1}{G} \sum_{i=1}^{G} \hat{r}_i \nabla_\theta \log \pi_\theta(Y^*(x) \mid x, z_i),$$

with the understanding that the multi-reference log-probability is induced by the summed event mass rather than by a single completion token sequence. The overall update would include the KL penalty:

$$\nabla_\theta J_{\mathrm{MRVF}} \approx g_z + g_r + g_{\mathrm{KL}}.$$

This formulation is the theoretical target that justifies the data pipeline built in the repository. The pipeline exists because $Y^*(x)$ must be constructed before $r_\theta(x, z, Y^*)$ can be estimated.

2.8 Research questions and evaluation protocol

The preliminary thesis is organized around four research questions:

1. Can a large joke corpus be transformed into weak prompts with multiple associated references rather than one-to-one joke pairs?

2. Does the current keyword-and-retrieval approximation produce prompt-reference sets that are large and coherent enough to support future MRVF training?

3. What failure modes are already visible in the pilot artifacts before the reinforcement-learning stage begins?

4. Which parts of the full training and evaluation procedure are already implemented, and which remain future work?

These questions imply a two-level evaluation protocol. The present thesis evaluates the pilot by descriptive statistics and qualitative analysis of prompt-reference sets. The final thesis should extend this with model-level comparisons across at least four dimensions inspired by recent humor-evaluation work [Sakabe et al., 2025, Hashemi et al., 2024, Gunjal et al., 2025]:

- prompt adherence;
- novelty;
- human-likeness;
- funniness.

For now, the methodological contribution is to make this later benchmark meaningful by first constructing the prompt-reference substrate on which it would depend.

Chapter 3

Results and Discussion

3.1 Overview of the implemented pilot

The main result of the current project stage is not a trained humor model. It is a working pilot pipeline that operationalizes the thesis idea at the data and evaluation levels. This result is more substantial than it may initially appear. In a non-verifiable domain, data construction is not a trivial preprocessing step; it is part of the method itself. If the data representation is wrong, the reinforcement-learning objective will optimize the wrong behavior.

The implemented pilot already demonstrates five important facts:

1. large-scale joke aggregation is feasible and already completed;
2. embedding-based retrieval can be used to build sets of references for weak prompts;
3. prompt construction can be approximated through keyword extraction;
4. pairwise generation and evaluation infrastructure is available for future model benchmarking;
5. the resulting artifacts expose concrete limitations that can guide the next implementation stage.

For a preliminary thesis, this is an acceptable and useful stopping point. It provides enough implemented material for a methodology chapter and enough empirical artifacts for a first results chapter, while still leaving room for a final thesis to deliver the actual reinforcement-learning experiments.

3.2 Pilot dataset statistics

Table 3.1 summarizes the key counts extracted from the local repository state. These numbers are especially important because they show the difference between the total available corpus, the current embedded retrieval pool, the seed set used for keyword extraction, and the final deduplicated prompt-reference dataset.

Table 3.1: Pilot data-construction statistics

Artifact	Count
Raw merged joke corpus	664,048
Embedded jokes in current local pilot	10,005
Keyword-annotated jokes in current pilot	1,005
Generated prompt-reference rows before deduplication	5,958
Unique prompt-reference rows after deduplication	5,058
Training prompts	4,046
Validation prompts	506
Test prompts	506

Several observations follow directly from these numbers.

First, the pipeline expands a relatively small seed subset into a much larger prompt-level dataset. From 1,005 keyword-annotated jokes, it produces 5,958 candidate prompt rows. This expansion is expected because each joke contributes multiple one-keyword and two-keyword combinations.

Second, deduplication removes 900 duplicate prompt rows. This is a useful sign rather than a problem. It indicates that different jokes frequently converge to the same weak prompt, which is exactly the condition needed for multi-reference learning.

Third, the size difference between the embedded retrieval pool (10,005 jokes) and the keyword seed set (1,005 jokes) reveals the current pilot strategy: a smaller set of seed jokes is used to generate prompts, but references are retrieved from a larger pool. This is methodologically sensible because it maximizes reference diversity without requiring the full prompt-construction pipeline to run over the entire corpus.

3.3 Reference-set quality

The most important question for the current thesis is whether the retrieved reference sets are large and coherent enough to justify a multi-reference objective. Table 3.2 reports descriptive statistics derived from the local prompt-reference artifacts.

Table 3.2: Descriptive statistics of deduplicated prompt-reference sets

Statistic	Value
Average references per prompt	4.07
Median references per prompt	5
Maximum references per prompt	5
Prompts with zero retrieved references	204
Prompts with exactly one reference	398
Prompts with five references	3,450
Prompts built from one keyword	2,117
Prompts built from two keywords	2,941
Average keyword-group size	1.58

These results are encouraging. The median prompt already reaches the configured reference cap of five items, and 3,450 out of 5,058 prompts are saturated at that limit. This indicates that the retrieval mechanism is not merely finding occasional neighbors; for a large fraction of prompts it constructs a dense local neighborhood of jokes that are semantically related enough to pass the similarity threshold.

At the same time, the long tail is visible. There are 204 prompts with zero retained references and 398 with only one reference. This matters for the later training stage because MRVF becomes weak or unstable when the prompt has no meaningful reference set. The current pilot therefore suggests a natural filtering or curriculum strategy: future training should probably begin with prompts that have at least two or three references and only later decide how to incorporate sparse cases.

The relative abundance of two-keyword prompts is also informative. In the deduplicated dataset, 2,941 prompts are built from two keywords and 2,117 from one keyword. This suggests that the current parameter choice (`min_keywords`=1, `max_keywords`=2) yields a balanced mixture of broad and moderately constrained

prompts. The one-keyword prompts increase diversity; the two-keyword prompts increase specificity.

3.4 Qualitative examples

Quantitative counts alone do not show whether the constructed prompts are semantically meaningful. For that reason, qualitative inspection of pilot rows is necessary. Table 3.3 provides representative examples drawn from the current training split.

Table 3.3: Examples of prompt-reference sets from the pilot dataset

Keyword group	Number of references	Illustrative reference behavior
`[alphabet, does]`	5	References cluster around alphabet-based wordplay, including pirate-letter jokes and orthographic misunderstandings. The prompt is weak, yet the reference set still shares a recognizably playful mechanism.
`[typo, ate]`	5	The references include jokes about autocorrect, eating, and lexical distortion. This is a good example of a prompt that is broad enough to admit multiple answers but still constrained by a recoverable comic theme.
`[ears]`	5	The retrieved references revolve around hearing or cornfield homophones. This shows that even a one-keyword prompt can generate a coherent neighborhood when the word carries a strong pun affordance.

Keyword group	Number of references	Illustrative reference behavior
`[cider, favourite]`	2	The prompt retrieves a much narrower set. This reveals the risk of under-supported prompts: they may be too idiosyncratic for robust training even if they remain semantically meaningful.
`[girl, lazy]`	1	The prompt effectively collapses to a near-singleton neighborhood. Such rows are useful for evaluation but weak for multi-reference reinforcement learning.

These examples support three conclusions.

First, the retrieval mechanism often preserves a plausible humor mechanism even though it is not explicitly told to search for mechanisms. This is a positive sign for the embedding-based approach.

Second, one-keyword prompts can still be useful when the keyword is semantically rich or strongly associated with punning. This means that prompt broadness is not automatically a weakness.

Third, sparsity remains a real issue. Some prompts are clearly under-supported, which means that future training cannot simply treat every prompt row as equally informative.

3.5 Interpretation of the pilot as a multi-reference dataset

The pilot results support the central methodological claim of the thesis: a joke corpus can indeed be restructured from isolated samples into weak prompts paired with multiple references. This is not merely a convenience for evaluation. It changes the shape of the learning problem.

In a standard SFT dataset, the model sees (x, y^*) pairs and is asked to imitate the observed answer. In the pilot dataset produced here, the learning

problem begins to look like $(x, Y^*(x))$, where the goal is not to reproduce one answer exactly, but to shift probability mass toward a set of acceptable answers. That is precisely the structure required by the MRVF objective.

The descriptive statistics also make the conceptual advantages of multi-reference modeling concrete. If the majority of prompts already have several references in a small pilot, then a larger-scale run over the full corpus has a reasonable chance of yielding a much richer multi-reference dataset. The pilot therefore functions as an existence proof for the data representation.

3.6 Limitations revealed by the current implementation

The pilot also reveals several limitations that must be acknowledged explicitly.

3.6.1 Prompt quality is approximate

The current system does not yet perform full prompt inversion from jokes. It uses keywords as a proxy. This is computationally efficient and easy to audit, but it inevitably loses structure. A keyword prompt such as `Write a joke using the following keyword(s): ears` can recover the topic of a joke, but not necessarily its framing, rhetorical setup, or intended comedic mechanism.

This limitation does not invalidate the pilot. It means that the current prompt representation should be interpreted as a *weak constraint*, not a rich semantic specification. That interpretation is consistent with the thesis goal: the training task should preserve multiplicity of valid outputs. Weak prompts may even be preferable for that reason, provided they remain coherent enough for retrieval.

3.6.2 Coverage is still partial

Only a subset of the total corpus has been embedded, and an even smaller subset has been used for keyword extraction in the current local run. Therefore, the reported numbers are pilot indicators, not final corpus-scale statistics. A final thesis should rerun the pipeline on a much larger portion of the corpus before

drawing strong conclusions about reference coverage.

3.6.3 Reference incompleteness remains unresolved

Even when the system finds five references, that does not mean the true answer set has been observed fully. Five is only the configured retrieval cap. The present pilot therefore confirms the relevance of the incompleteness problem discussed in the methodology chapter. The retrieved set is useful, but it should still be interpreted as a partial observation of a larger latent set of good jokes.

3.6.4 No committed benchmark outputs yet

The repository contains implemented candidate-generation and evaluation modules, but the current committed data artifacts do not yet include candidate generations or pairwise evaluation results. This means the present thesis can defend the benchmark design, but not yet report final model rankings. That limitation is acceptable in a preliminary thesis, provided it is stated clearly.

3.6.5 Training remains future work

The most important limitation is also the most obvious one: the actual reinforcement-learning stage has not yet been implemented. The theoretical objective is specified, and the data pipeline now justifies it, but a final thesis must still test whether MRVF training improves humor generation relative to base, instruct, and reasoning checkpoints.

3.7 What the pilot already contributes

Despite those limitations, the current work already contributes something non-trivial.

First, it replaces a purely conceptual report with a thesis-grade research narrative grounded in actual code and actual data artifacts. The methodology is no longer hypothetical in the abstract; major components are implemented and inspectable.

Second, it demonstrates that the move from single-reference humor supervision to multi-reference supervision is practically achievable. This is arguably the main precondition for the later training study.

Third, it clarifies the software architecture needed for the final stage. The project now has separate modules for corpus ingestion, representation learning, prompt construction, retrieval, generation, and evaluation. This modularity reduces the risk that future negative results will be uninterpretable because all components were entangled.

Fourth, it identifies the exact frontier between what has already been validated and what still requires experimentation. That clarity is itself valuable in a thesis context. It prevents overclaiming and allows the final version of the thesis to focus its experimental effort where uncertainty is highest.

3.8 Next stage of the study

Based on the current pilot, the next stage of the project should proceed in the following order.

First, the prompt-construction pipeline should be rerun at a larger scale so that the distribution of reference counts becomes more stable. A final thesis should not rely on only 1,005 keyword-seeded jokes if the full corpus is available.

Second, sparse prompts should be filtered or stratified. Prompts with zero or one retrieved reference are useful diagnostic cases, but they are poor anchors for multi-reference reward estimation.

Third, the `training.py` module should implement the MRVF update with careful variance control and KL regularization.

Fourth, model comparisons should be run over at least four checkpoint families: base, instruction-tuned, reasoning-tuned, and MRVF-trained variants. This matches the experimental sketch of the original report but relocates it to the correct stage of the project.

Fifth, automatic pairwise evaluation should be complemented with a smaller but carefully designed human annotation study. The literature strongly suggests that humor cannot be left entirely to model judges [Sakabe et al., 2025, Zhang et al., 2024].

In short, the pilot has done the preparatory work needed to make a fi-

nal reinforcement-learning experiment worthwhile. It has not yet replaced that experiment.

Conclusion

This thesis addressed humor generation as a non-verifiable reasoning problem. The central claim was that training for humor should move away from single-reference imitation and toward multi-reference probability allocation over weak prompts. The literature review showed why this shift is needed: SFT is prone to imitation and memorization, preference-based RL is unstable in humor due to noisy judgments, and current multistep humor systems improve quality at the cost of hard-coded reasoning structures.

The work made two connected contributions. At the methodological level, it formulated a multi-reference extension of verifier-free reinforcement learning in which the reward is the total probability mass assigned to a set of acceptable references rather than to one gold answer. At the implementation level, it built and analyzed a pilot pipeline that turns a large joke corpus into prompt-reference sets through embeddings, keyword extraction, vector retrieval, deduplication, and data splitting.

The empirical analysis of the pilot showed that the idea is viable. In the current local repository state, a corpus of 664,048 jokes has already been assembled, a pilot subset has been embedded and processed, and the resulting workflow produces 5,058 deduplicated prompt-reference rows with an average of 4.07 references per prompt and a median of five. These statistics indicate that the core structural requirement of MRVF learning, namely the existence of prompts with multiple acceptable references, can be satisfied in practice.

At the same time, the thesis remains preliminary in the correct sense. The prompt-construction procedure is still approximate, the downstream benchmark artifacts are not yet committed, and the MRVF training loop itself has not yet been implemented. Therefore, the present work should be read as a thesis foundation rather than as a final empirical verdict on reinforcement learning for humor.

The main conclusion is that the project has now crossed the boundary from a speculative report to a defensible research program. The data representation,

software pipeline, and formal objective are aligned. The next step is to implement the training stage and test whether the constructed multi-reference dataset can indeed elicit better reasoning and stronger humor than standard post-training baselines.

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Appendix A

Implemented Prompt Templates

This appendix records the main templates already implemented in the repository. They are included here because they define the operational form of the prompts used throughout the pilot.

A.1 Reference-generation prompt

The prompt used for reference retrieval and candidate generation is:

```
Write a joke using the following keyword(s): {keywords}
```

Its main advantage is consistency. The same textual form is used for retrieval-time prompt embedding and for later candidate generation, which reduces mismatch between dataset construction and evaluation. Its main disadvantage is that it is intentionally generic and therefore does not encode humor theory or audience information.

A.2 Embedding query instructions

Two short instruction templates are used to adapt the embedding model to retrieval subtasks:

```
Instruct: Given a keyword for a joke, retrieve jokes
would match the keyword.
Query: {keyword}
```

```
Instruct: Given a prompt for a joke, retrieve jokes
that would answer the prompt.
Query: {prompt}
```

These templates are important because the pilot does not use raw keyword strings as embeddings directly. It embeds them inside retrieval-oriented instructions, which makes the semantic intent of the vector space more task-specific.

A.3 Evaluation prompt

The pairwise evaluation system prompt asks the judge to compare two jokes written for the same prompt and choose exactly one winner. The instruction explicitly discourages rewarding length or polish when these do not improve humor. This design is intentionally simple and should be viewed as a baseline evaluation protocol rather than a final human-aligned rubric.

Appendix B

Supplementary Notes on the Software Architecture

The repository is organized as a set of asynchronous pipelines:

- `JokesPipeline`: downloads and preprocesses raw corpora;
- `EmbeddingsPipeline`: computes dense joke embeddings;
- `KeywordsPipeline`: extracts salient keywords by similarity and MMR;
- `ReferencesPipeline`: builds prompt-reference datasets through vector retrieval;
- `CandidatesPipeline`: generates jokes from prompts;
- `EvaluationPipeline`: performs pairwise judging and leaderboard aggregation.

This architecture matters for the final thesis because it separates data construction from model comparison. As a result, the later MRVF training stage can be added without rewriting the rest of the infrastructure. The missing component is precisely the one that now becomes the main target of future work: a training pipeline that consumes the constructed prompt-reference dataset and optimizes the formal objective introduced in Chapter 2.